Final Research Report

Dynamic Bayesian Persuasion with partial commitment

Hanshuo Ma

McMaster University

Department of Economics

Instructor: Prof. Gajendran Raveendranathan

ECON 701

July 30, 2026

Contents

1 Introduction

In this paper, the state of the economy is in a dynamic and recurrent change between normal and recession - a Business Cycle. And the goal of the Central Bank - the information designer, is to find an optimal information policy such that it stimulates spending as much as possible throughout the Business Cycle. However, the central bank can only *partially* commit the optimal policy as they will face political pressure to pursue manipulation of *FX* rate through monetary policy regardless of the state of the Business Cycle. The household is aware of this partial commitment of the information policy of the central bank and they will only want to increase spending when their belief of the current state as normal is above certain probability.

Concretely, suppose the economy has two states, Normal and Recession, and the household's utility depends on both the state and their own action; the household wants to spend in the Normal state and save in a Recession. However, the household does not observe the state directly; they learn about it only through the central bank's signaling scheme, which takes the form of interest-rate guidance issued each period. The bank's own payoff, by contrast, is independent of the state; it depends only on the household's action, specifically on spending, which is what gives the bank an incentive to signal a cut regulation regardless of the true state. The trouble is that a cut signal is not free for the bank to send; whenever the bank cuts guidance during a Recession, it also artificially deflates the *exchange rate* in the global market, strengthening the country's exports while making the household relatively less wealthy in global terms. Suppose the household travels abroad annually on vacation; every such artificial cut erodes the purchasing power of that trip. This is precisely the channel through which short-run political pressure operates on the bank. If the household believed the central bank committed fully to its stated policy, they would expand consumption whenever the guidance said Normal and cut back whenever it said Recession. But once the household suspects the bank is only partially committed, the guidance loses some of its credibility, and the household may prefer to save even when told the state is Normal.

1.1 Motivation

The main reason the Bayesian persuasion model stood out from the traditional communication protocol such as cheap talk (Crawford & Sobel 1982), and signaling games (Spence 1973) is that Bayesian persuasion endows the sender with more commitment power. However, in real life most senders or information designers (i.e. an algorithm) are responsible for multiple mandates. For instance, central bank have a dual mandates of minimize inflation and maximizing employment; both mandates often require deviation from one another within the context of monetary economics. Moreover, providing a more extreme scenario - the Central Bank put under political pressure to change or signaling to change their monetary policy in favour of short-term growth during a period of uncertain economic outlook. Hence, the assumption of full commitment of central bank's signaling scheme need to be relaxed from time to time, and to do so within a *dynamic Bayesian persuasion* model rather than a static one-shot canonical model allowed the model to simulate the dynamic of business cycle: recessions arrive, persist, and end, and the bank must issue guidance again and again over the same cycle.

1.2 Research question

In a dynamic persuasion model with exogenous partial commitment and a recurrent Markov process, what is optimal policy? Is it delaying disclosure while risk is low and then revealing a state change only as often as obedience requires?

1.3 Approach

The main model follows the dynamic Bayesian persuasion literature (Ely 2017) closely apart from a few important distinction. Instead of two states where one is absorbing, in this paper no state is absorbing. Therefore, good behavoiour is no longer front-loaded: the household no longer changes behavior exactly once, from working to shirking in Ely (2017)[1] Partial commitment set up follows

[1]In Ely (2017) the employee receive beeps while working, indicating they can check their email - entering the shirking state. The shirking state is terminal, hence the goal of the manager - the information designer is to design an

Min (2021), the announced signal is honoured with probability λ; with probability $1-\lambda$ the bank re-optimises (cheats).

1.4 Related literature and contribution

The canonical static Bayesian Persuasion model of Kamenica and Gentzkow (2011) consists of one sender and one receiver. The receiver must take an action, and the receiver's utility from that action is state-dependent; depending only on the action the receiver takes. The sender's goal is to design an information policy that persuades the receiver into the sender-preferred action as often as possible, regardless of which state is realized. The defining assumption is commitment: the sender commits to a signaling scheme before the state realizes, and the scheme is fixed and publicly known to the receiver. Upon observing the signal realization, the receiver updates beliefs via Bayes' rule. The induced distribution of posteriors must satisfy Bayes plausibility: it must average back to the prior belief. The optimal scheme is then read off a concavification: writing the sender's payoff as a function of the receiver's posterior, the sender's value is the concave envelope of that function evaluated at the prior, and the optimal signal splits the prior into the posteriors that support the envelope.

There four main extensions of the canonical static model of Kamenica and Gentzkow (2011): (a) multiple receivers, (b) multiple senders, (c) dynamic environments, and (d) sender's commitment. A key discipline in this literature is that the receiver knows the degree of commitment, with probability $1-\lambda$ the principal[2] may alter the signal realization, but because the receiver interprets every message through this lens, the extra flexibility generates no equilibrium advantage — the sender's ex-ante payoff is weakly increasing in λ (Min 2021). If credibility falls too far, the receiver rationally disregards all communication and acts on the prior alone, reducing the sender to the no-communication payoff.

This project sits at the intersection of the dynamic Bayesian Persuasion with full commitment

information policy such that it maximizes the duration of the amount of time the employee stays in the working state. The optimal policy from the paper is to beep with a delay

[2]From now on in this paper I will refer to information designer, sender, and in this paper's specific context, the central bank as the principle.

from Ely (2017) and the static Bayesian Persuasion with partial commitment from Min (2021), so far, have mostly been studied separately, each relaxing the canonical model in one direction.

2 One-Sender, One-Receiver Model in Discrete Time

2.1 Model setup

Suppose that the household have a threashold p^* such that she will start saving as soon as she believes with probability greater than p^* that a recession is happening. The household enters period t with beliefs μ_t. The principal then observes the current state s_t, draw their own commitment type and sends a message to the household accordingly. In response to the message the household updates to an interim belief v_t and takes an action. All of the preceding can be regarded as happening instantaneously. Next, a discrete length of time passes and we enter period $t+1$.

The household's choice of action a_t maximizes the expected value of her state-dependent utility function $x, a_t \in \arg\max_a E_{v_t} x(a,s)$. The principal's payoff $u(a,s)$ indirectly depends on the household's interim belief, i.e., $u(v,s) = u(a(v),s)$. The long-run average discounted payoff of the principal is then $(1-\delta)\sum_{t=0}^{\infty} \delta^t u(v_t, s_t)$, where $\delta \in (0,1)$ is the discrete-time discount factor.

A policy for the principal is a rule $\sigma(h_t) \in \Delta M_t$ which maps the principal's complete prior history h_t into a probability distribution over messages, where the messages space M_t is unrestricted.

Messages as interest-rate guidance. The message space is left unrestricted — the obfuscation principle below handles arbitrary message spaces, and no reduction is imposed. On the path of the optimal policy, however, only two messages are ever used, and we name them by the instrument they represent: CUT, the reassuring guidance that keeps the household spending, and RAISE, an increase in the policy rate, which is the disclosure event following Ely (2017). The RAISE is an informative signal: on the optimal path it's issued only in the recession state, so it reveals the state outright. A period in which the announced rule issues CUT in both states with probability one is uninformative: the guidance is uncorrelated with the state and beliefs move by drift alone. The same CUT message that carries no information in the delay phase becomes informative in the rationing phase of Section 4, where it's good news: conditional on CUT, the posterior returns to exactly p^* (equation (15)).

2.2 Principal's payoff function

The household wishes to match her action to the state. Specifically, she strictly prefers action $a = 1$ when $v > p^*$, she strictly prefers action $a = 0$ when $v < p^*$, and she is indifferent whenever $v = p^*$. On the other hand, the principal has state-independent preferences and always strictly prefers the household to choose $a = 0$. We will normalize the principal's flow payoffs so that he earns 1 whenever the household chooses action $a = 0$ and 0 otherwise. Because the household's action will be determined by her belief v_t, we can collapse the principal's flow payoff to a function that depends directly on the interim belief v:

$$u(v) = \begin{cases} 1 & v \leq p* \\ 0 & v > p* \end{cases}$$

Notation. Throughout, μ_t denotes the belief entering period t and ν_t the interim belief after the period t message, so that $\mu_{t+1} = f(\nu_t)$; under uninformative guidance $\nu_t = \mu_t$. The belief μ_t is public: it's pinned down by the common prior μ_0, the announced rule, and the history of messages, all of which both parties observe. The three phases of the optimal policy in Section 4 are not three separate commitments: the government commits once, at time zero, to a single state- and belief-contingent rule, and the phases are simply the regions of $[0,1]$ that the computable belief path visits.

2.3 Finding the Belief map

In this basic model, there are two states $s_t \in S = \{N, R\}$, and the household has two actions $a \in \{0, 1\}$. The process is a recurrent two-state Markov chain with the following transition matrix:

$$P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$$

$a = \Pr(s_{t+1} = R|s_t = N), b = \Pr(s_{t+1} = N|s_t = R)$, where $a, b \in (0,1), a > b, a+b < 1$. The set up here is to make the recession state (R) more "sticky" in stationary distribution μ^*. Solving $\pi P = \pi$ with $\pi_N + \pi_R = 1$ gives $\pi_N a = \pi_R b$, hence

$$\mu^* = (\pi_N, \pi_R) = \left(\frac{b}{a+b}, \frac{a}{a+b}\right) = (1-\mu_R^*, \mu_R^*), \qquad \mu_R^* = \frac{a}{a+b}$$

Because $a > b$ we have $\mu_R^* > \frac{1}{2}$; the recession state is entered faster than it is left, so beliefs rest above one half.

Here we are more interested in the case where $\mu_R^* > p^*$, as in the long run, if the stationary belief satisfied $\mu_R^* \leq p^*$, the principal would simply leave the household be and let them keep spending forever. Under this non-degeneracy condition, the interval $I = (p^*, 1]$ is an absorbing interval under the belief map f defined below, and the flow payoff $u \equiv 0$ is (trivially) linear on I; therefore, using Lemma 1 from Ely (2017, Online Appendix 6.2), the value function V must be affine on I.

Lemma 1 From Ely (2017 Online Appendix 6.2).

Suppose that I is absorbing under f, and that u is linear over I. Then V is also linear on I. Here V is the value function of the dynamic problem; it's the concavification of a function that involves the value function itself, extending the static persuasion of Kamenica and Gentzkow (2011).

Definition (Belief map).

The Markov chain induces a belief-transition map $f : \Delta(\Theta) \to \Delta(\Theta)$. If today's belief is μ then the receiver's belief at the start of next period is the one-step prediction $f(\mu) = P^\top \mu$.

$$P^\top \mu = \begin{pmatrix} 1-a & b \\ a & 1-b \end{pmatrix} \begin{pmatrix} \mu_N \\ \mu_R \end{pmatrix} = \begin{pmatrix} (1-a)\mu_N + b\,\mu_R \\ a\,\mu_N + (1-b)\,\mu_R \end{pmatrix}.$$

Substitute $\mu_N = 1-\mu_R = 1-\mu$:

$$\begin{aligned} a(1-\mu)+(1-b)\mu &= a - a\mu + \mu - b\mu \\ &= a + \mu\,(1-a-b) \\ &= a + (1-a-b)\,\mu \;=\; f(\mu). \end{aligned}$$

The map f is affine with slope $1-a-b \in (0,1)$, and its unique fixed point solves $\mu = a + (1-a-b)\mu$, i.e. $\mu = \frac{a}{a+b} = \mu_R^*$. The deterministic belief drift and the stochastic chain share the same rest point, which confirms the stationary distribution above. To verify the absorbing-interval claim: $f(p^*) - p^* = a - (a+b)p^* = (a+b)(\mu_R^* - p^*) > 0$ and $f(1) = 1-b < 1$, so f maps $(p^*, 1]$ into itself.

Remark (the four roles of the belief map). Since the household acts on beliefs rather than on states as she never observes s_t. The entire two-state chain P is compressed into this one affine map on the one-dimensional simplex.

- *(i) f is the clock.* Under uninformative guidance $\nu_t = \mu_t$, so the belief drifts deterministically along f toward μ_R^*; starting below the threshold, the drift (13) crosses p^* at the hitting time (14), which fixes the length of the delay phase of the optimal policy (Section 4).

- *(ii) f links the periods.* In the recursion (2) the future enters only through the composition $V \circ f$; without it the dynamic problem would collapse into a sequence of static concavifications.

- *(iii) f at the threshold forces disclosure.* Because $\varphi = f(p^*) > p^*$, guidance that carries no information pushes the belief past the threshold into the no-rollover region, so holding the investor at p^* requires the informative RAISE to be rationed at exactly the rate ρ^* in (15).

- *(iv) f at the top delivers recurrence and the credibility bound.* $f(1) = 1-b < 1$: certainty is not a fixed point of the belief map, so a RAISE is non-terminal and $V(1) > 0$ (Section 3.2); the same number $1-b$ is then the largest on-path belief at which rationing operates, so the commitment threshold of Section 5 is $\lambda_{\text{dyn}} = \rho^*(1-b)$, records $f(1) < 1$ as the belief-dynamics expression of recurrence

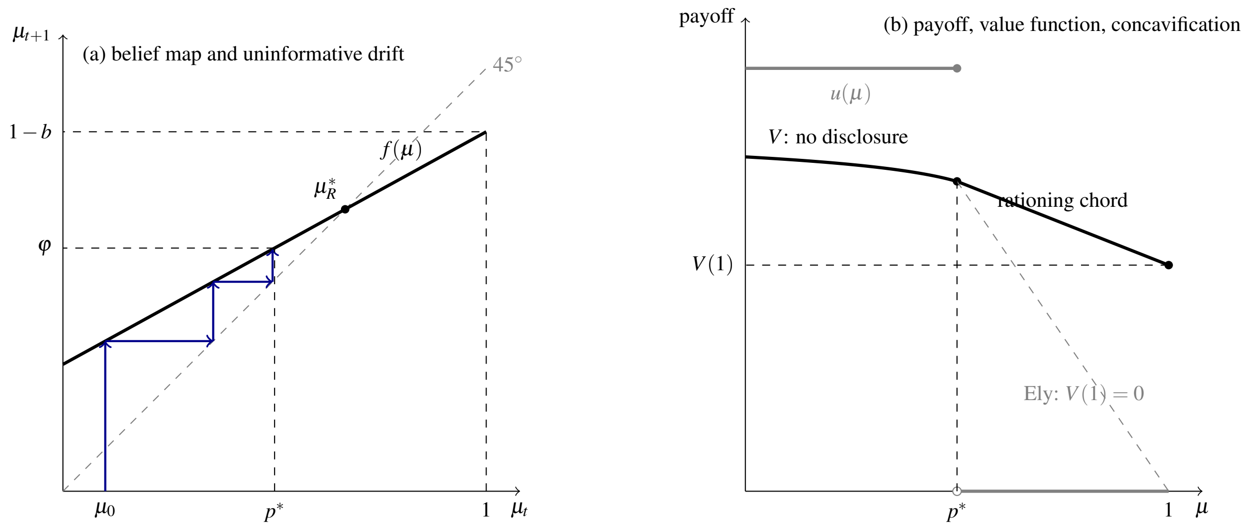

Figure 1: The geometry of the solution on the one-dimensional simplex. *Panel (a):* the affine belief map $f(\mu) = a + (1-a-b)\mu$, its fixed point μ_R^* (also the stationary raise probability), and the uninformative-guidance drift from μ_0 cobwebbing up to the threshold p^*; the images $\varphi = f(p^*)$ and $1-b = f(1)$ bound the absorbing interval. *Panel (b):* the step flow payoff u (grey), the concave value function V on the no-disclosure region $[0, p^*]$, and the affine rationing chord joining $(p^*, V(p^*))$ to $(1, V(1))$ — the kink at p^* is where disclosure begins. The dashed grey chord is Ely's absorbing benchmark, where $V(1) = 0$; the vertical gap at $\mu = 1$ is the recurrence premium. Drawn to scale for $a = 0.3$, $b = 0.15$, $p^* = 0.5$, $\delta = 0.9$ using the closed forms of Section 3.2: $V(p^*) = 0.733$, $V(1) = 0.535$.

3 Full-commitment benchmark

Before relaxing commitment we solve the principal's problem under full commitment, which both fixes the benchmark payoff and pins down the optimal disclosure policy that partial commitment must later sustain as shown in Min (2021).

3.1 The Bellman equation

Now in order to obtain the value function, we need Obfuscation Principle from Ely (2017) to fully characterize the dynaic optimization problem.

Lemma 2(The Obfuscation Principle):. Any policy σ induces a stochastic process (μ_t, v_t) satisfying

1. $E(\mu_t|v_t) = \mu_t$

2. There exists an linear map f, s.t. $\mu_{t+1} = f(v_t)$

The principal's expected payoff from such a policy is

$$E\sum_{t=0}^{\infty}\delta^t u(v_t)$$

With Lemma 2, we can state our Bellman equation:

$$V(\mu_t) = \max_{q\in\Delta(\Delta S), Eq=\mu_t} E_q[(1-\delta)u(v_t) + \delta(V\circ f)(v_t)] \tag{1}$$

$$V = \text{cav}[(1-\delta)u + \delta(V\circ f)] \tag{2}$$

Here cav denotes concavification from Kamenica and Gentzkow (2011). However, contrast to the static persuasion problem, the difficulty is that the function being concavificated here appears on both sides of the equation. Thus going back to the Lemma 1, since we have already have shown that the *interval* $I = (p^*, 1]$, is absorbing and u is linear there, the value function V is affine on I. Now, restricting attention to the region I, the real-valued affine function $V(\mu) = \vec{a}\mu + \vec{b}$ on$(p^*, 1]$ is completely determined by its values at exactly two break points $V(p^*)$, and $V(1)$.

$$V(p^*) = \vec{a}p^* + \vec{b} \tag{3}$$

$$V(1) = \vec{a} + \vec{b} \tag{4}$$

$$V(1) - V(p^*) = \vec{a}(1-p^*)$$

$$\vec{a} = \frac{V(1) - V(p^*)}{1-p^*} \tag{5}$$

Now once we have $\vec{a}$, we can plug this back in equation (4) and solve for $\vec{b}$:

$$\begin{aligned}
\vec{b} &= V(1) - \vec{a} \\
\vec{b} &= V(1) - \frac{V(1) - V(p^*)}{1 - p^*} \\
\vec{b} &= \frac{V(p^*) - p^* V(1)}{1 - p^*}
\end{aligned} \tag{6}$$

Finally, the affine function $V(\mu) = \vec{a}\mu + \vec{b}$ has the form:

$$V(\mu) = \frac{V(1) - V(p^*)}{1 - p^*}\mu + \frac{V(p^*) - V(1)p^*}{1 - p^*} \qquad \forall \mu \in [p^*, 1] \tag{7}$$

It is convenient to record the equivalent "chord" form of (7), obtained by collecting the $V(p^*)$ and $V(1)$ terms:

$$V(\mu) = \frac{(1-\mu)V(p^*) + (\mu - p^*)V(1)}{1 - p^*}, \qquad \mu \in [p^*, 1], \tag{8}$$

which exhibits $V(\mu)$ as the Bayes-plausible convex combination of the two endpoint values.

Now we may look at everything more closely and evaluate the function at $\mu = 1$ and $\mu = p^*$. We know that $V \circ f(1) = V(f(1))$ where $f(1) = a + (1 - a - b) = 1 - b$. Similarly, according to the principal's payoff, $(1-\delta)u(1) = 0$. A household certain of a recession cannot be persuaded otherwise, so every policy at $\mu = 1$ is outcome-equivalent to uninformative cut guidance, the bank receives the flow payoff of 0 and the belief propagates to $f(1) = 1 - b$. With both terms known, and evaluating the chord (8) at $\mu = 1 - b \in (p^*, 1)$, equation (2) gives:

$$V(1) = \delta V(1-b) = \delta\, \frac{bV(p^*) + (1 - b - p^*)V(1)}{1 - p^*} \tag{9}$$

Taking similar steps at the threshold, where the household still spends so the flow is $(1-\delta)u(p^*) = 1 - \delta$, and where silence propagates the belief to

$$\varphi := f(p^*) = a + (1 - a - b)p^* \in (p^*, 1)$$

we also obtain:

$$V(p^*) = (1-\delta) + \delta V(\varphi) = (1-\delta) + \delta\,\frac{(1-\varphi)V(p^*) + (\varphi - p^*)V(1)}{1-p^*} \tag{10}$$

again using the chord (8), now at $\mu = \varphi$. Equations (9) and (10) are two linear equations in the two unknowns $V(p^*)$ and $V(1)$, and we now solve them explicitly.

3.2 Solving the two-equation system

Multiply (9) through by $1-p^*$ and collect the $V(1)$ terms:

$$\begin{aligned}
(1-p^*)V(1) &= \delta b V(p^*) + \delta(1-b-p^*)V(1) \\
\big[(1-p^*) - \delta\big((1-p^*)-b\big)\big]V(1) &= \delta b V(p^*) \\
\big[(1-p^*)(1-\delta) + \delta b\big]V(1) &= \delta b V(p^*)
\end{aligned}$$

Define $q := 1-p^*$ and $\Delta := q(1-\delta) + \delta b > 0$,

$$\boxed{V(1) = \frac{\delta b}{\Delta} V(p^*)}, \qquad \Delta = (1-p^*)(1-\delta) + \delta b \tag{11}$$

Multiply (10) through by $q = 1-p^*$:

$$qV(p^*) = (1-\delta)q + \delta(1-\varphi)V(p^*) + \delta(\varphi - p^*)V(1)$$

Substituting (11) for $V(1)$,

$$qV(p^*) - \delta(1-\varphi)V(p^*) - \frac{\delta^2 b(\varphi - p^*)}{\Delta}V(p^*) = (1-\delta)q$$

so that

$$\boxed{V(p^*) = \frac{(1-\delta)(1-p^*)}{(1-p^*) - \delta(1-\varphi) - \dfrac{\delta^2 b(\varphi - p^*)}{\Delta}}}, \qquad \varphi - p^* = (a+b)(\mu_R^* - p^*) > 0 \qquad (12)$$

Equations (11) and (12), together with the (12), deliver V in closed form on all of $[p^*, 1]$.

Consistency checks.

- *(i) The denominator of* (12) *is positive and* $V(p^*) \leq 1$. Since $\Delta > \delta b$, the last term satisfies $\frac{\delta^2 b(\varphi - p^*)}{\Delta} < \delta(\varphi - p^*)$; hence

$$(1-p^*) - \delta(1-\varphi) - \frac{\delta^2 b(\varphi - p^*)}{\Delta} > (1-p^*) - \delta(1-\varphi) - \delta(\varphi - p^*) = (1-p^*)(1-\delta) > 0,$$

  which simultaneously shows $V(p^*) < 1$; the central bank cannot do better than the household spending in every period, which the normalized utility payoff is 1.

- *(ii)* $0 < V(1) < V(p^*)$. Immediate from (11) since $0 < \frac{\delta b}{\Delta} < 1$; the value is higher at the threshold than at certainty of recession, as it must be, and V is decreasing along the chord.

- *(iii) The absorbing limit recovers Ely.* As $b \to 0$ the recession never ends and we should recover Ely's absorbing beeps, where reaching certainty is terminal. Indeed $\Delta \to q(1-\delta) \neq 0$ while the numerator of (11) carries the factor $b \to 0$, so $V(1) \to 0$. For $b > 0$ we have $V(1) > 0$, reaching certainty of recession is no longer terminal, because the chain eventually leaves R and the household resumes spending.

The pair V(1) and V(p^*) suffices to describe the entire solution because the optimal policy only ever induces two posteriors, p^* (keep spending) and 1 (confession); the whole dynamic problem reduces to valuing these two destinations, and every other value in the disclosure region is their convex combination along the chord.

4 Optimal disclosure policy

The optimal policy is read off the concave envelope in the manner of Ely (2017): disclose the cut guidance where the envelope coincides with the inner objective, and randomize (disclose) where the envelope strictly exceeds it. This gives a three-phase belief path.

Phase A — uninformative cut guidance ($\mu \le p^*$). While the belief is below the threshold the central bank issues cut guidance - regardless of the state, collecting the flow payoff 1 each period while the belief drifts. Subtracting the fixed-point identity $\mu_R^* = a + (1-a-b)\mu_R^*$ from $\mu_{t+1} = a + (1-a-b)\mu_t$ and iterating from μ_0,

$$\mu_t = \mu_R^* + (1-a-b)^t\,(\mu_0 - \mu_R^*), \tag{13}$$

so the belief converges monotonically toward μ_R^*. Starting from $\mu_0 < p^* < \mu_R^*$, setting $\mu_{t^*} = p^*$ and solving,

$$t^* = \frac{\ln \dfrac{\mu_R^* - p^*}{\mu_R^* - \mu_0}}{\ln(1-a-b)} > 0, \tag{14}$$

and in discrete time the belief reaches or just crosses p^* at the integer period $T(\mu_0) = \lceil t^* \rceil$. This is the recurrent analogue of Ely's initial silence, only the drift constant $1-a-b$ and the resting point μ_R^* have changed.

Phase B — rationing at the threshold. Once μ reaches p^*, the no-information drift would carry it to $f(p^*) = \varphi > p^*$, into the region where the household stops spending. To hold the belief on the threshold, the bank discloses a recession signal - an informative raise guidance *only* in state R, at rate $\rho \in [0,1]$, and never in state N. The single number ρ is the *disclosure rate*: $\rho = 1$ is full confession, $\rho = 0$ is stonewalling, and the RAISE guidance sent in state R (probability $1-\rho$) is a deliberate false negative. After the drift the pre-disclosure belief is some $\mu > p^*$; conditioning on the event of a cut guidance and requiring the posterior to equal p^*:

$$\Pr(R \mid \text{CUT}) = \frac{\mu(1-\rho)}{\mu(1-\rho)+(1-\mu)\cdot 1} \overset{!}{=} p^*.$$

Cross-multiplying and solving for ρ:

$$\begin{aligned}
\mu(1-\rho) &= p^*\left[\mu(1-\rho)+(1-\mu)\right] \\
\mu(1-\rho)(1-p^*) &= p^*(1-\mu) \\
1-\rho = \frac{p^*(1-\mu)}{\mu(1-p^*)} &\implies \rho = \frac{\mu(1-p^*)-p^*(1-\mu)}{\mu(1-p^*)},
\end{aligned}$$

hence the threshold-holding disclosure rate is

$$\boxed{\rho^*(\mu) = \frac{\mu - p^*}{\mu\,(1-p^*)}} \tag{15}$$

This is the "false-negative" message that pins the belief on the obedience boundary, conditional on the cut guidance, the household is just willing to keep spending.

Phase C — the recurrent cycle. When a raise guidance is issued, the message reveals state R, so the posterior jumps to $\mu = 1$. In Ely's absorbing model this would be terminal. In the recurrent chain it is not, from $\mu = 1$ the drift pulls the belief back down through $f(1) = 1-b$ toward μ_R^*, the belief re-enters the rationing region, and the cycle repeats. This is why $V(1) > 0$ in (11), the bank periodically gets the household expanding again.

Proposition 1: Optimal full-commitment policy.

Under the non-degeneracy condition $p^* < \mu_R^*$, the optimal full-commitment policy is "delay-then-ration", stay silent while $\mu \le p^*$ (Phase A), and once $\mu = p^*$ issue a raise guidance only in state R, at the rate $\rho^*(\mu)$ in (15) that holds the post-cut guidance belief at p^* (Phase B). Disclosure is non-terminal, after a raise guidance the belief drifts down from 1 and the cycle repeats (Phase C).

This answers the research question in the affirmative for the full-commitment case, the optimum does delay disclosure while recession risk is low, then reveals a state change only as often as obedience ($\mu \leq p^*$) requires.

5 Partial commitment

We now relax commitment using the stochastic-commitment formulation of Min (2021) and Lipnowski, Ravid, and Shishkin (2022), each period the announced disclosure rule is honoured with probability $\lambda \in [0,1]$, and with probability $1-\lambda$ the bank re-optimizes (cheats) given the current state. The question becomes when the delay-then-ration policy of Proposition 1 survives period by period.

The credibility constraint. The temptation to cheat is largest exactly when the bank is supposed to raise the guidance, a raise makes the household stop spending today, which is precisely the outcome the bank dislikes in the short run. With honouring probability λ, the *realized* raise rate in state R is $\lambda\rho$ rather than the promised ρ. For the household to remain obedient — to keep the post-cut belief at or below p^* — the realized disclosure must be at least the required rate at the relevant belief:

$$\lambda\rho \;\geq\; \rho^*(\mu) \quad \text{at the on-path belief } \mu.$$

Since the bank can promise at most $\rho = 1$ (raise with certainty in R), credibility requires $\lambda \geq \rho^*(\mu)$ at every on-path belief where rationing operates, hence at the *largest* such belief.

The binding belief. From (15), $\partial\rho^*/\partial\mu = \frac{p^*}{\mu^2(1-p^*)} > 0$, so the worst case is the largest belief at which the bank still rations. After a raise guidance the belief is 1 and then drifts to $f(1) = 1-b$; this is the highest pre-disclosure belief the rationing phase must contend with on the cycle. Evaluating (15) there gives the dynamic commitment threshold.

Proposition 2: Sustainability of persuasion.

The delay-then-ration policy is sustainable under partial commitment if and only if the honouring probability satisfies

$$\boxed{\lambda \;\geq\; \lambda_{\text{dyn}} := \rho^*(1-b) = \frac{1-b-p^*}{(1-b)(1-p^*)}} \tag{16}$$

The threshold is well posed because $p^* < \mu_R^* < 1-b$ (the latter since $\frac{a}{a+b} \leq 1-b \iff 0 \leq b(1-a-b)$).

Proposition 3: Persistence, not arrival, drives the commitment need.

λ_{dyn} is independent of the onset rate a, strictly decreasing in the threshold p^*, and strictly decreasing in the exit rate b. Equivalently, the commitment requirement is governed by the expected recession duration $1/b$, longer-lasting recessions demand more commitment, while how often recessions arrive is irrelevant.

6 Report summary and beyond

The full-commitment benchmark is now solved in closed form (equations (11) and (12)), the optimal policy is characterized (Proposition 1), and the partial-commitment layer delivers a single sufficient statistic λ_{dyn} for whether persuasion is sustainable (Proposition 2), driven by recession *persistence* rather than *arrival* (Proposition 3).

Two main extensions remain for the final report. First, *endogenous commitment*: here λ is exogenous (Min 2021); the goal stated in the research question is to make it an equilibrium object, e.g. a reputation state updated after observed cheats as in Best & Quigley (2024). Second, *behavior below the cutoff*: characterizing the optimal (necessarily coarser) disclosure when $\lambda < \lambda_{\text{dyn}}$.

References

[1] Arieli, I. and Stewart, C. (2025). Bayesian Persuasion without Commitment. Working paper, arXiv:2511.18662.

[2] Best, J. and Quigley, D. (2024). Persuasion for the Long Run.

[3] Crawford, V. and Sobel, J. (1982). Strategic Information Transmission. *Econometrica* 50(6), 1431–1451.

[4] Danilov, V. I. and Lambert-Mogiliansky, A. (2018a). Targeting in Quantum Persuasion Problem. *Journal of Mathematical Economics* 78, 142–149.

[5] Danilov, V. I. and Lambert-Mogiliansky, A. (2018b). Preparing a (Quantum) Belief System. *Theoretical Computer Science* 752, 97–103.

[6] Ely, J. (2017). Beeps. *American Economic Review* 107(1), 31–53.

[7] Kamenica, E. and Gentzkow, M. (2011). Bayesian Persuasion. *American Economic Review* 101(6), 2590–2615.

[8] Lipnowski, E., Ravid, D., and Shishkin, D. (2022). Persuasion via Weak Institutions. *Journal of Political Economy* 130(10), 2705–2730.

[9] Mailath, G. and Samuelson, L. (2006). *Repeated Games and Reputations: Long-Run Relationships*. Oxford University Press.

[10] Min, D. (2021). Bayesian Persuasion under Partial Commitment. *Economic Theory*.

[11] Spence, M. (1973). Job Market Signaling. *Quarterly Journal of Economics* 87(3), 355–374.